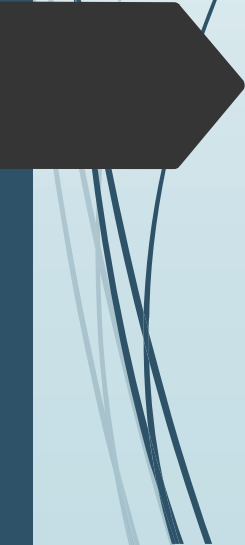


Introduction

Chapter 1.



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Preliminary Remarks



Section 1.1

Section Summary

- Significant Digits of Precision: Examples
- Errors: Absolute and Relative
- Accuracy and Precision
- Rounding and Chopping
- Nested Multiplication
- First Programming Experiment
- Mathematical Software
- Summary

Preliminary Remarks

- Many students have calculators and access to mathematical software systems that can produce solutions to complicated numerical problems with minimal difficulty.
- The purpose of a numerical mathematics course is to examine the underlying algorithmic techniques so that students learn how the software or calculator found the answer. In this way, they would have a better understanding of the inherent limits on the accuracy that must be anticipated in working with such systems.

Preliminary Remarks

- One of the fundamental strategies behind many of numerical methods is the replacement of a difficult problem with a string of simpler ones.
- This strategy succeeds in finding zeros of functions (Chapter 3), interpolation (Chapter 4), numerical integration (Chapters 5–6), and solving linear systems (Chapters 7–8).

Significant Digits of Precision:

- **Significant digits** are digits beginning with the leftmost *nonzero* digit and ending with the rightmost *correct* digit, including final zeros that are exact.
- To show the effect of the number of significant digits used in a calculation, we consider the problem of solving a linear system of equations.

Significant Digits of Precision:

- **Example 1:** Let us concentrate on solving for the variable y in this linear system of equations in two variables :

$$0.1036 x + 0.2122 y = 0.7381$$

$$0.2081 x + 0.4247 y = 0.9327 \quad (1)$$

- **First**, carry only three significant digits of precision in the calculations.
- **Second**, repeat with four significant digits throughout.
- **Finally**, use ten significant digits.

Significant Digits of Precision:

➤ **First** we round all numbers in the original problem to three digits and round all the calculations, keeping only three significant digits.

➤ We take a multiple $\sigma = 0.208/0.104 \approx 2.00$.

➤ The new coefficient of the x-term is :

$$0.208 - (2.00)(0.104) \approx 0.208 - 0.208 = 0$$

➤ And the new y-term coefficient is:

$$0.425 - (2.00)(0.212) \approx 0.425 - 0.424 = 0.001.$$

➤ The right-hand side is :

$$0.933 - (2.00)(0.738) = 0.933 - 1.48 = -0.547.$$

➤ Hence, we find that :

$$y = -0.547/(0.001) \approx -547.$$

Significant Digits of Precision:

➤ **Second**, We decide to keep four significant digits throughout and repeat the calculations.

➤ We take $\sigma = 0.2081/0.1036 \approx 2.009$.

➤ The new coefficient of the x-term is :

$$0.2081 - (2.009)(0.1036) \approx 0.2081 - 0.2081 = 0,$$

➤ And the new y-term coefficient is:

$$0.4247 - (2.009)(0.2122) \approx$$

$$0.4247 - 0.4263 = -0.001600,$$

➤ The right-hand side is :

$$0.9327 - (2.009)(0.7381) \approx$$

$$0.9327 - 1.483 \approx -0.5503.$$

➤ Hence, we find that :

$$y = -0.5503/(-0.001600) \approx 343.9.$$

Significant Digits of Precision:

- ▶ We find that the **answer** has changed from -547 to 343.9 , which is a huge difference!.
- ▶ In fact, if we **repeat** this process and carry ten significant decimal digits, we find that even 343.9 is not accurate, since we obtain $356.29071\ 99$.
- ▶ ***The lesson learned in this example is that data thought to be accurate should be carried with full precision and not be rounded off prior to each of the calculations.***

Errors: Absolute and Relative?

► Why we measuring errors?

- 1) To determine the accuracy of numerical results.
- 2) To develop stopping criteria for iterative algorithms.

► **Errors** : Suppose that α and β are two numbers, of which one is regarded as an approximation to the other.

► The **error** of β as an approximation to α is $\alpha - \beta$; that is, the error equals the exact value minus the approximate value.

► The **absolute error** of β as an approximation to α is :
 $|\alpha - \beta|$.

► The **relative error** of β as an approximation to α is :
 $|\alpha - \beta| / |\alpha|$

Errors: Absolute and Relative?

- In summary, we have :

$$\text{absolute error} = |\text{exact value} - \text{approximate value}|$$

$$\text{relative error} = \frac{|\text{exact value} - \text{approximate value}|}{|\text{exact value}|}$$

- **Example2:** Consider $x = 0.00347$ rounded to $\underline{x} = 0.0035$ and $y = 30.158$ rounded to $\underline{y} = 30.16$. In each case, what are the number of significant digits, absolute errors, and relative errors. Interpret the results.

Solution : Case 1. $\underline{x} = 0.35 \times 10^{-2}$ has two significant digits, absolute error 0.3×10^{-4} , and relative error 0.865×10^{-2} . Case 2. $\underline{y} = 0.3016 \times 10^2$ has 4 significant digits, absolute error 0.2×10^{-2} , and relative error 0.66×10^{-4} .

Clearly, the relative error is a better indication of the number of significant digits than the absolute error.

Accuracy and Precision :

- ▶ Accurate to n decimal places means that you can trust n digits to the right of the decimal place.
- ▶ Accurate to n significant digits means that you can trust a total of n digits as being meaningful beginning with the leftmost nonzero digit.
- ▶ Suppose you use a ruler graduated in millimeters to measure lengths. Then :
 - ▶ One millimeter = 0.001 m. Accurate to 3 decimal places.
 - ▶ 12.3456789 m would be meaningless, it should be 12.345 m or 12.346 m.
 - ▶ 12.345 m Accurate to 3 decimal places, 5 significant digits.

Accuracy and Precision :

- In **adding and subtracting** numbers, the result is accurate only to the smallest number of significant digits used in any step of the calculation. For Example, the result of :

$$(1.2) + (3.45) = 4.6$$

actually has only two significant digits of accuracy.

- In **multiplication and division**, perform these computations on a calculator:

$$(1.23)(4.5) = 5.535 \text{ and } (1.23)/(4.5) = 0.27333\ 3333.$$

You think that there are four and nine significant digits in the results, but there are really only two!

- **Conclusion :?**

As a rule of thumb, one should keep as many significant digits in a sequence of calculations as there are in the least accurate number involved in the computations.

Rounding and Chopping :

- We say that a number x is **chopped to n digits** or figures when all digits that follow the n th digit are discarded and none of the remaining n digits are changed. Conversely, x is **rounded to n digits** or figures when x is replaced by an n -digit number that approximates x with minimum error.
- **For Example ,**
Rounding some three-decimal numbers to two digits are :
 $0.217 \approx 0.22$, $0.365 \approx 0.37$, $0.475 \approx 0.48$, and $0.592 \approx 0.59$,
While chopping them gives :
 $0.217 \approx 0.21$, $0.365 \approx 0.36$, $0.475 \approx 0.47$, and $0.592 \approx 0.59$.
- **Conclusion :?**
On the computer, the user sometimes has the option to have all arithmetic operations done with either chopping or rounding. The latter is usually preferable, of course.

Nested Multiplication?

- We will begin with some remarks on evaluating a polynomial efficiently and on rounding and chopping real numbers.
- To evaluate the polynomial :

$$p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_{n-1}x^{n-1} + a_nx^n$$

- We group the terms in a nested multiplication:

$$p(x) = a_0 + x(a_1 + x(a_2 + \cdots + x(a_{n-1} + x(a_n)) \cdots))$$

- The pseudocode that evaluates $p(x)$ can be:

```
integer  $i, n$ ;   real  $p, x$ ;   real array  $(a_i)_{0:n}$   
 $p \leftarrow a_n$   
for  $i = n - 1$  to  $0$  do  
     $p \leftarrow a_i + xp$   
end for
```

Nested Multiplication?

- This nested multiplication procedure is also known as Horner's algorithm or synthetic division.
- *For example,*

$$p(x) = 5 + 3x - 7x^2 + 2x^3$$

should be computed as

$$p(x) = 5 + x(3 + x(-7 + x(2)))$$

for a given value of x .

Clearly, We have avoided all the exponentiation operations by using nested multiplication!

Nested Multiplication?

- Horner's algorithm can be used in the **deflation** of a polynomial. This is the process of removing a linear factor from a polynomial. If r is a root of the polynomial p , then $x - r$ is a factor of p . The remaining roots of p are the $n - 1$ roots of a polynomial q of degree 1 less than the degree of p such that:

$$p(x) = (x - r)q(x) + p(r)$$

Where : $q(x) = b_0 + b_1x + b_2x^2 + \dots + b_{n-1}x^{n-1}$

- The pseudocode for Horner's algorithm can be written as follows:

```

integer  $i, n$ ;   real  $p, r$ ;   real array  $(a_i)_{0:n}, (b_i)_{0:n-1}$ 
 $b_{n-1} \leftarrow a_n$ 
for  $i = n - 1$  to 0 do
     $b_{i-1} \leftarrow a_i + r b_i$ 
end for
  
```

- Notice that $b_{-1} = p(r)$ in this pseudocode.

Nested Multiplication?

- If the calculation in Horner's algorithm is to be carried out with pencil and paper, the following arrangement is often used:

$$\begin{array}{cccccc}
 & a_n & a_{n-1} & a_{n-2} & \dots & a_1 & a_0 \\
 r) & & r b_{n-1} & r b_{n-2} & \dots & r b_1 & r b_0 \\
 \hline
 & b_{n-1} & b_{n-2} & b_{n-3} & \dots & b_0 & b_{-1}
 \end{array}$$

- **Example3:** Use Horner's algorithm to evaluate $p(3)$, where p is the polynomial:

$$p(x) = x^4 - 4x^3 + 7x^2 - 5x - 2$$

- We arrange the calculation as suggested above:

Nested Multiplication?

$$\begin{array}{r} 1 \quad -4 \quad 7 \quad -5 \quad -2 \\ 3) \quad \quad \quad 3 \quad -3 \quad 12 \quad 21 \\ \hline 1 \quad -1 \quad 4 \quad 7 \quad 19 \end{array}$$

- Thus, we obtain $p(3) = 19$, and we can write :

$$p(x) = (x - 3)(x^3 - x^2 + 4x + 7) + 19$$

- **Example4** : Deflate the polynomial p of the preceding example, using the fact that 2 is one of its zeros. We use the same arrangement of computations as explained previously:

$$\begin{array}{r} 1 \quad -4 \quad 7 \quad -5 \quad -2 \\ 2) \quad \quad \quad 2 \quad -4 \quad 6 \quad 2 \\ \hline 1 \quad -2 \quad 3 \quad 1 \quad 0 \end{array}$$

- Thus, we have $p(2) = 0$, and

$$x^4 - 4x^3 + 7x^2 - 5x - 2 = (x - 2)(x^3 - 2x^2 + 3x + 1)$$

First Programming Experiment?

- This paragraph is a Homework to increase the experience in programming using numerical methods. **“Refer to textbook.”**

Summary

- (1) Use **nested multiplication** to evaluate a polynomial efficiently:

$$\begin{aligned} p(x) &= a_0 + a_1x + a_2x^2 + \cdots + a_{n-1}x^{n-1} + a_nx^n \\ &= a_0 + x(a_1 + x(a_2 + \cdots + x(a_{n-1} + x(a_n)) \cdots)) \end{aligned}$$

A segment of pseudocode for doing this is:

```
p ← an
for k = 1 to n do
    p ← xp + an-k
end for
```

- (2) **Deflation** of the polynomial $p(x)$ is removing a linear factor:

Where:

$$p(x) = (x - r)q(x) + p(r)$$

$$q(x) = b_0 + b_1x + b_2x^2 + \cdots + b_{n-1}x^{n-1}$$

Summary

- The pseudocode for Horner's algorithm for **deflation** of a polynomial is:

```

$$b_{n-1} \leftarrow a_n$$
for  $i = n - 1$  to  $0$  do  
     $b_{i-1} \leftarrow a_i + r b_i$   
end for
```

- Here $b_{-1} = p(r)$.

END